SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	**CSA1003**	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	**22/07/2026**	**Duration:**	60 minutes

Code of Conduct

I **Y. Ramadevi – 192373065** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Y. Ramadevi

Title: AI-Based Trip Planner and Itinerary Optimization System

1. Understand the Problem Statement

a) Explain the problem in your own words

Planning a trip involves searching for destinations, comparing transportation options, booking accommodations, managing budgets, scheduling activities, and handling unexpected changes such as weather or traffic. Most travelers use multiple websites and applications, making the planning process time-consuming and inefficient.

An AI-Based Trip Planner and Itinerary Optimization System is designed to simplify travel planning by using Artificial Intelligence to generate personalized travel itineraries based on user preferences, budget, travel duration, weather conditions, and real-time traffic information. The system automatically recommends destinations, optimizes travel routes, and updates the itinerary whenever changes occur.

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b) Objectives of the Proposed System

- Develop an intelligent trip planning platform.
- Generate personalized travel itineraries using AI.
- Optimize travel routes to minimize travel time and cost.
- Recommend hotels, restaurants, attractions, and transportation.
- Provide real-time updates for weather, traffic, and travel delays.
- Improve user convenience and travel experience.
- Reduce manual planning effort.

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c) Scope of the Problem

The system covers:

- User registration and login
- Destination search
- AI-based itinerary generation
- Budget planning
- Hotel and transport recommendations
- Route optimization
- Weather forecasting integration
- Map and GPS navigation
- Trip modification
- Booking management
- Travel history
- Feedback and ratings

The system does not directly provide transportation or hotel services but integrates with third-party travel APIs.

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2. Identify Key Stakeholders and Challenges

a) Stakeholders

1. Travelers/Users
2. Travel Agencies
3. Hotel Providers
4. Transport Providers
5. Tourism Department
6. System Administrator
7. Payment Gateway Provider
8. API Service Providers (Maps, Weather, Booking)

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b) Roles and Responsibilities

Stakeholder	Roles and Responsibilities
Travelers	Enter preferences, view itinerary, book trips, provide feedback
Travel Agencies	Manage travel packages and offers
Hotel Providers	Update room availability and pricing
Transport Providers	Provide transport schedules and fare information
Tourism Department	Update tourist information and attractions
Administrator	Manage users, monitor system performance, maintain data
Payment Gateway	Secure online payment processing
API Providers	Provide maps, weather forecasts, booking services

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c) Major Challenges

Travelers

- Time-consuming planning
- Budget management
- Route optimization
- Finding reliable information

Travel Agencies

- Managing customer requests
- Personalized recommendations

Hotel Providers

- Dynamic pricing
- Availability updates

Administrator

- System security
- Data accuracy
- High server load

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3. Analyze the Current Approach

a) Existing System

Currently, travelers use several independent platforms:

- Google Maps
- Hotel booking websites
- Flight booking portals
- Weather applications
- Travel blogs
- Tourism websites

The itinerary is usually created manually.

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b) Strengths

- Large amount of travel information
- Easy online booking
- GPS navigation available
- Online reviews

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c) Limitations

- Multiple applications required
- Manual itinerary preparation
- No intelligent recommendations
- Difficult budget management
- No automatic route optimization
- Limited personalization

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d) Problems/Gaps

- Time-consuming planning
- Duplicate information search
- Lack of centralized platform
- Frequent itinerary changes
- Poor optimization of travel schedules

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4. Proposed Software Solution Using Software Engineering Principles

a) Proposed Solution

Develop an AI-powered web/mobile application that automatically creates optimized travel itineraries using:

- Artificial Intelligence
- Machine Learning
- GPS Navigation
- Weather APIs
- Map APIs
- Travel Recommendation Engine

The system recommends:

- Best destinations
- Hotels
- Restaurants
- Tourist attractions
- Transportation
- Optimized travel schedule

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b) Major Functional Modules

1. User Management

- Registration
- Login
- Profile Management

2. Destination Recommendation

- AI-based destination suggestions
- Trending locations

3. Trip Planning Module

- Travel date selection

- Budget planning
- Interest selection

4. Itinerary Optimization

- AI scheduling
- Route optimization
- Distance calculation

5. Booking Module

- Hotels
- Flights
- Buses
- Cabs

6. Navigation Module

- Google Maps integration
- Live traffic updates

7. Weather Monitoring

- Weather prediction
- Alerts

8. Expense Tracker

- Budget estimation
- Daily expenses

9. Feedback Module

- Ratings
- Reviews

10. Admin Module

- User management
- Data management
- Reports

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c) Functional Requirements

- User Registration/Login
- Search Destinations
- Generate AI Itinerary

- Optimize Routes
- Hotel Recommendation
- Booking Integration
- Budget Estimation
- Weather Updates
- GPS Navigation
- Notifications
- Feedback Submission
- Admin Dashboard

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d) Non-Functional Requirements

Performance

- Fast itinerary generation
- Response time below 3 seconds

Security

- User authentication
- Data encryption
- Secure payments

Reliability

- High system availability (99.9%)
- Backup and recovery

Scalability

- Support thousands of concurrent users

Maintainability

- Modular architecture
- Easy updates

Usability

- User-friendly interface
- Responsive design

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e) Software Engineering Principles Used

Modularity

Separate modules for booking, itinerary, navigation, and recommendations.

Scalability

Cloud-based architecture to support increasing users.

Maintainability

Reusable components and clean coding practices.

Reliability

Automatic backups and error handling.

Usability

Simple dashboards and intuitive navigation.

Security

- Role-based access control
- Secure authentication
- Encrypted user data
- Secure API communication

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5. Justify the Chosen Methodology/Framework

Selected SDLC Model: Agile (Scrum)

Why Agile is Suitable

The AI-Based Trip Planner requires:

- Frequent feature updates
- Continuous testing
- User feedback incorporation
- API integrations
- AI model improvements

Agile allows development in small iterations, enabling quick adaptation to changing user needs and technological advancements.

Advantages of Agile

- Continuous customer feedback
- Faster delivery
- Easy requirement changes
- Incremental development
- Better software quality
- Frequent testing
- Reduced project risk

How Agile Supports Development

- Sprint planning
- Daily Scrum meetings
- Sprint reviews
- Continuous integration
- Continuous testing
- Frequent releases
- Easier maintenance

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6. Expected Outcomes and Limitations

a) Expected Benefits

- Intelligent travel planning
- Personalized recommendations
- Reduced planning time
- Better budget management
- Optimized travel routes
- Improved customer satisfaction
- Real-time itinerary updates
- Centralized travel platform

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b) How the Solution Addresses Challenges

Challenge	Proposed Solution
Manual planning	AI-generated itinerary
Route optimization	AI route planning
Budget issues	Expense estimation
Multiple applications	Single integrated platform
Travel delays	Real-time updates
Lack of personalization	AI recommendation engine

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c) Risks and Limitations

Risks

- API service failures
- Internet connectivity issues
- Cybersecurity threats
- Incorrect AI recommendations
- High cloud infrastructure costs

Limitations

- Depends on external APIs
- Requires internet connectivity
- AI accuracy depends on available data
- Booking services rely on third-party providers

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d) Future Enhancements

- Voice-enabled travel assistant
- AI chatbot for travel support
- Multilingual support
- Augmented Reality (AR) tourist guide
- Offline itinerary access
- Predictive travel demand analysis
- Carbon footprint estimation
- Integration with wearable devices
- Smart travel alerts using IoT
- Blockchain-based secure booking records

Conclusion

The AI-Based Trip Planner and Itinerary Optimization System provides a smart, personalized, and efficient solution for modern travel planning. By integrating Artificial Intelligence, route optimization, weather forecasting, booking services, and real-time updates into a single platform, the system minimizes manual effort and enhances the overall travel experience. Applying software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security ensures that the system is robust, flexible, and capable of supporting future enhancements. The Agile (Scrum) methodology further enables continuous improvement and rapid adaptation to evolving user needs, making the proposed solution both practical and sustainable.

ANNEXURE A – CASE STUDY ANALYSIS

AI-Based Trip Planner and Itinerary Optimization System

1. UNDERSTAND THE PROBLEM STATEMENT

Explain the problem

Trip planning is time-consuming and involves multiple platforms. There is no intelligent system to create personalized itineraries and optimize travel.

Objectives

- Develop an intelligent trip planning platform
- Generate personalized itineraries using AI
- Optimize routes to save time and cost
- Recommend hotels, restaurants, attractions
- Provide real-time updates
- Improve user convenience and experience

Scope

Covers user management, destination search, AI itinerary generation, booking, route optimization, weather updates, navigation, budget tracking, feedback and admin management.

2. IDENTIFY KEY STAKEHOLDERS & CHALLENGES

Key Stakeholders

- Travelers/Users
- Travel Agencies
- Hotel Providers
- Transport Providers
- Tourism Department
- System Administrator
- Payment Gateway Provider
- API Service Providers (Maps, Weather, Booking)

Roles & Responsibilities

- Travelers: Plan trips, book, give feedback
- Agencies: Manage packages and offers
- Hotels: Update availability and pricing
- Transport: Provide schedules and fares
- Tourism Dept: Update attractions info
- Admin: Manage users and system
- Payment Gateway: Secure transactions
- API Providers: Provide maps, weather, booking services

Major Challenges

- Time-consuming planning
- Budget management
- Route optimization
- Reliable information
- Dynamic pricing & availability
- System security & data accuracy

3. ANALYZE THE CURRENT APPROACH

Existing Approach

Users rely on multiple independent platforms like Google Maps, hotel booking sites, flight portals, weather apps, travel blogs, etc. Itinerary is prepared manually.

Strengths

- Large information availability
- Easy online booking
- GPS navigation available
- Online reviews

Limitations

- Multiple apps required
- Manual itinerary preparation
- No intelligent recommendations
- Difficult budget management
- No automatic route optimization
- Limited personalization

Gaps

- Time-consuming planning
- Duplicate information search
- Lack of centralized platform
- Frequent itinerary changes
- Poor optimization of schedules

4. PROPOSED SOLUTION USING SOFTWARE ENGINEERING PRINCIPLES

Proposed Solution

An AI-powered web/mobile application that creates optimized travel itineraries using AI, ML, GPS, Weather APIs, Map APIs and recommendation engine.

Major Functional Modules

1. User Management
2. Destination Recommendation
3. Trip Planning
4. Itinerary Optimization
5. Booking Module
6. Navigation Module
7. Weather Monitoring
8. Expense Tracker
9. Feedback Module
10. Admin Module

Functional Requirements

- User Registration/Login
- Search Destinations
- Generate AI Itinerary
- Optimize Routes
- Hotel Recommendation
- Booking Integration
- Budget Estimation
- Weather Updates
- GPS Navigation
- Notifications
- Feedback Submission
- Admin Dashboard

Non-Functional Requirements

- Performance
- Reliability
- Scalability
- Security
- Maintainability
- Usability

Software Engineering Principles

Modularity · Scalability · Maintainability · Reliability · Usability · Security

5. JUSTIFY THE CHOSEN METHODOLOGY / FRAMEWORK

Selected SDLC Model: Agile (Scrum)

SCRUM

Why Agile is Suitable?

The system requires frequent updates, user feedback, continuous testing, API integrations and AI model improvements. Agile allows quick adaptation to changing requirements.

Advantages of Agile

- Continuous customer feedback
- Faster delivery
- Easy requirement changes
- Incremental development
- Better software quality
- Frequent testing
- Reduced project risk

How Agile Supports Development

- Sprint planning
- Daily Scrum meetings
- Sprint reviews
- Continuous integration
- Continuous testing
- Frequent releases
- Easier maintenance

6. EXPECTED OUTCOMES AND LIMITATIONS

Expected Benefits

- Intelligent travel planning
- Personalized recommendations
- Reduced planning time
- Better budget management
- Optimized travel routes
- Improved customer satisfaction
- Real-time itinerary updates
- Centralized travel platform

How Solution Addresses Challenges

Challenge	Proposed Solution
Manual planning	AI-generated itinerary
Route optimization	AI route planning
Budget issues	Expense estimation
Multiple applications	Single integrated platform
Travel delays	Real-time updates
Lack of personalization	AI recommendation engine

Risks & Limitations

Risks: API failures, Internet issues, security threats, incorrect AI suggestions, high cloud costs.

Limitations: Depends on external APIs, requires internet, AI accuracy based on data, third-party booking dependency.

Future Enhancements

- Voice-enabled travel assistant
- AI chatbot for travel support
- Multilingual support
- AR tourist guide
- Offline itinerary access
- Predictive travel demand analysis
- Carbon footprint estimation
- Blockchain-based booking records